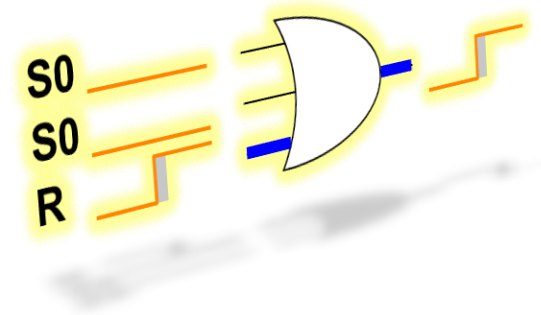


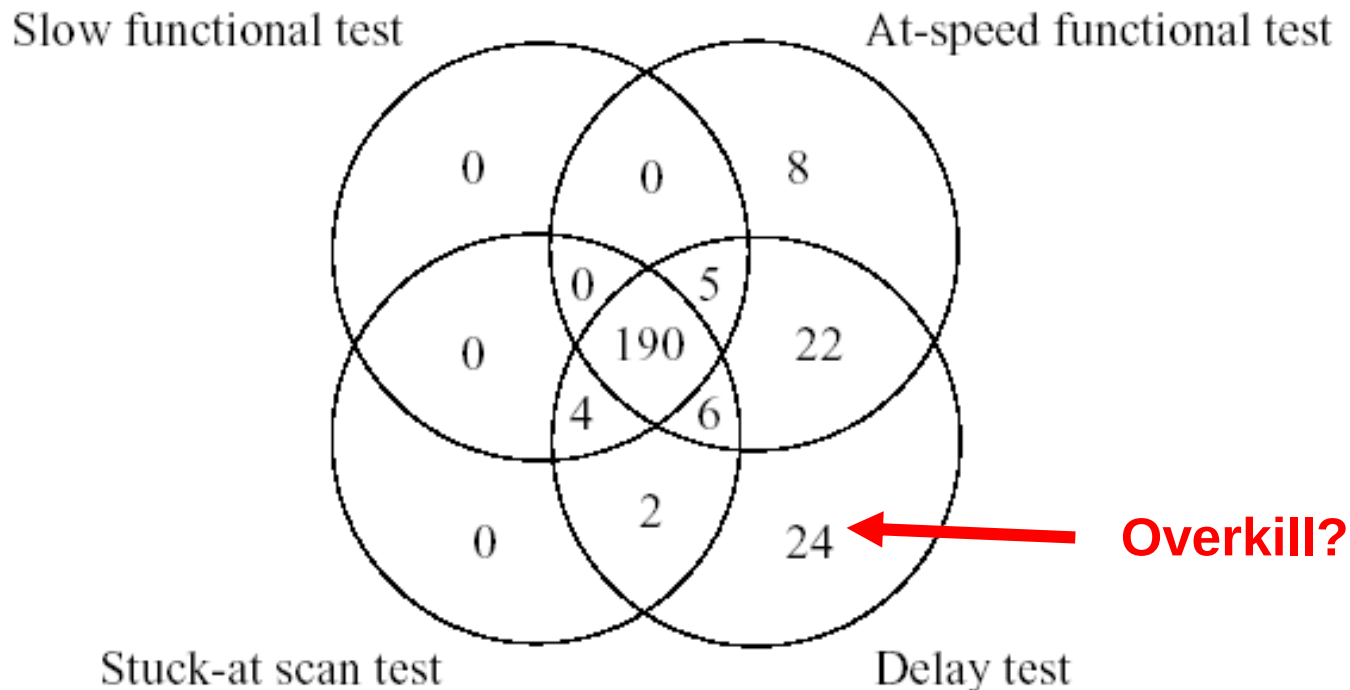
Delay Test

- Introduction and delay fault models
- Path Delay Fault
- Transition Delay Fault
- **Experimental Results* (not in exam)**
- **Issues of Delay Tests*** (not in exam)
- **Conclusions**



HP Experiment [Maxwell 96]

- Delay testing = TDF + some PDF
- Delay testing detects detected unique CUTs
 - ♦ that escaped other testing

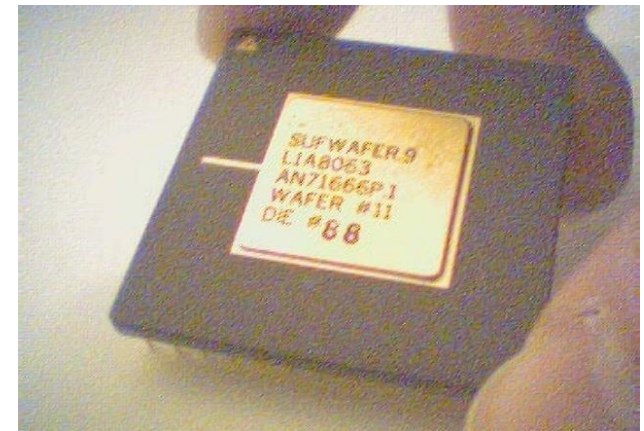


HP Experiment (Cont'd)

- **Observations: How useful are path delay tests?**
 - ◆ 2,000 longest paths chosen and only 148 paths have robust tests
 - ◆ Path delay fault tests only detect 18.5% of devices with delay faults

Stanford Experiment [EJM 00]

- Center for Reliable Computing, Stanford
- Murphy Experiment
 - ♦ Goal: evaluate effectiveness of different test techniques
- Murphy Chip
 - ♦ LSI Logic 150k CMOS Gate Array
 - ♦ 25k Gates, all combinational CUTs
 - ♦ 120-Pin Ceramic PGA Package
 - ♦ $L_{\text{eff}} = 0.7\mu\text{m}$
 - ♦ 5 volt Nominal Supply Voltage
 - ♦ Total 5.5K tested
 - * 116 defective chips found



SSF Test Results

				Escapes		
SSF Tool	Fault model	Cove- rage	Test length	characte rized	slow (1/3)	slow (1/30)
1**	gate	100	427	4	7	9
2*	gate	100	313	3	5	7
3**	pin	100	857	3	6	7
	gate	100	1,000	2	5	7
4**	pin	100	456	6	8	9
	gate	100	603	2	4	5
5**	pin	100	466	6	9	9
	gate	100	571	4	6	8
6*	gate	100	547	3	7	7
	gate	99	532	4	8	8
	gate	98	490	6	9	10
	gate	95	467	10	12	13
	gate	90	427	15	17	18
	gate	80	334	20	23	23
Weighted random*	gate	100	62,921	2	7	
Weighted random**	gate	100	21,892	2	7	
Exhaustive				0	2	4

* academic ATPG tools

** commercial ATPG tools

Non-SSF Test Results

- delay fault testing does not detect unique defective chips
 - ◆ Performance not particular impressive

Tool	Fault model	Test length	Escapes***		# of CUTs tested
			Characterized	Slow (1/3)	
	IDDQ	100	15		116
	VLV		3	4	116
1**	Transition	1,444	6	8	116
2*	Stuck open	1,610	4	9	116
3*	Gate delay fault		9	11	78§
	Gate delay fault, X->0		8	9	78§
4*	Robust path delay, X->0		5	5	60§
	Robust path delay, X->random		5	5	60§
5*	Robust path delay		3	5	60§
6**	Non-robust path delay		8	8	60§
	Non-robust path delay		2	5	60§
7	Verification		6	7	56§

* academic ATPG tools ** commercial ATPG tools

*** Tests applied at characterized speed

§ These patterns were not available for all of the CUT

Delay Test

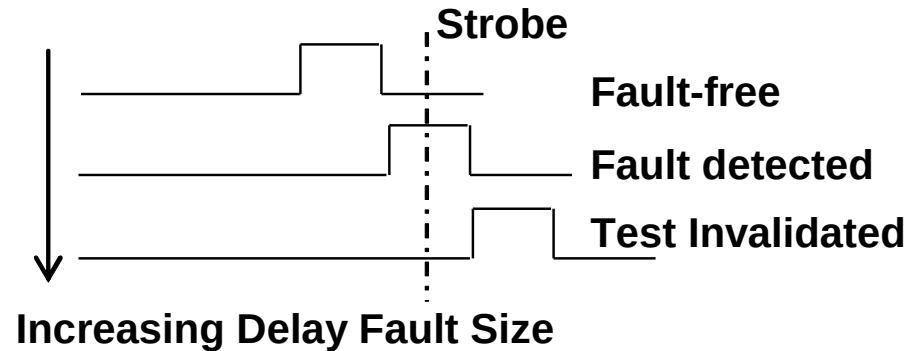
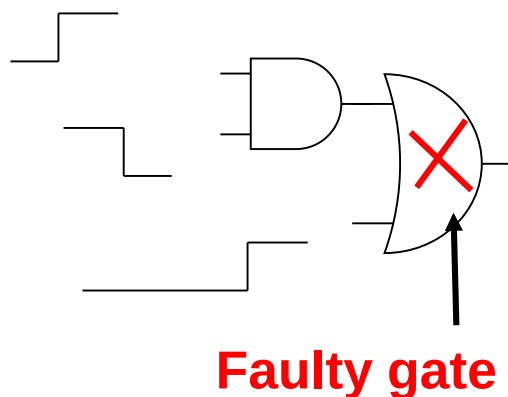
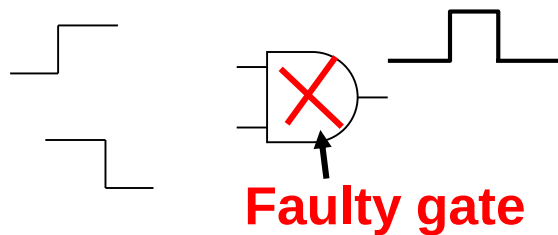
- Introduction and delay fault models
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Issues of Delay Tests

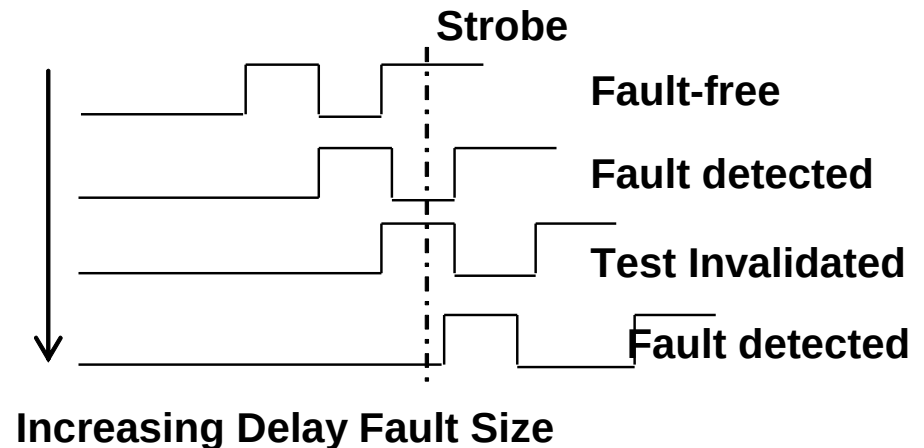
- Path selection is difficult
 - ◆ Too many paths in real circuits
 - ◆ e.g. 2×10^{20} paths in 32-bit multiplier (ISCAS'85 benchmark C6288)
 - ◆ Usually choose timing critical paths
 - * good to verify the speed, but not good for defect testing
- Need fast ATE to detect small delay faults
- Strobe time hard to determine
- Timing-aware ATPG is slow
- Many untestable paths
- Multi-cycle paths

Strobe Time Hard to Determine

- Strobe time hard to determine in advance
 - ♦ Fault size dependent
 - ♦ Process variation
- Example:

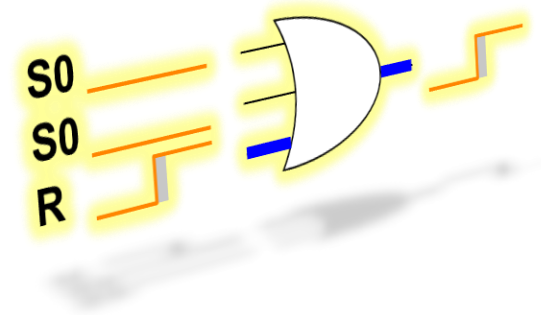


(a)

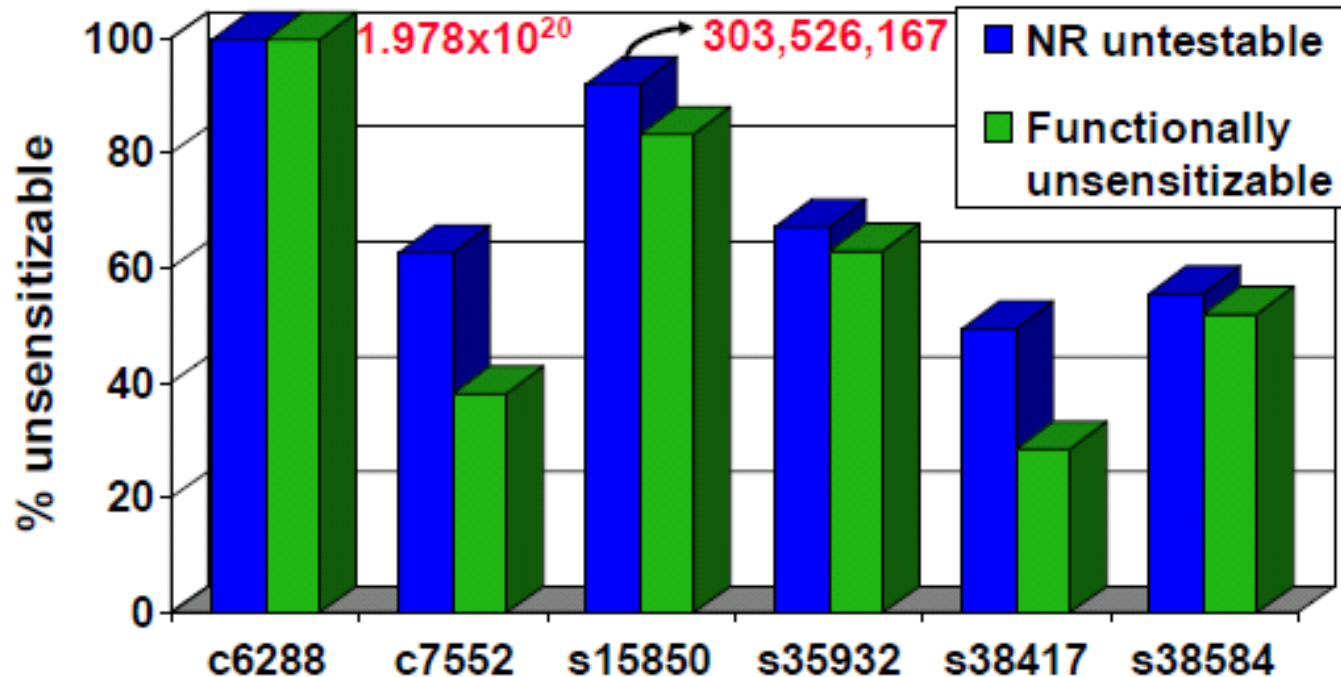


Delay Test

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Many Untestable Paths



Difficulties in Delay Tests

- At-speed testing needs expensive ATE
- Path list hard to generate
 - ◆ Too many paths
- *Small delay faults* occur more frequent in nano-meter technologies
 - ◆ Need propagate fault through long paths
 - * Timing-aware ATPG is slow
- At-speed delay testing very difficult
 - ◆ How to apply high speed clock in test mode?
 - ◆ At-speed delay testing induces signal/power integrity problem
 - * IR drop, ground bounce, x-talk noise ...
- Overkill
 - ◆ Some IC fail delay testing but actually work in system
 - ◆ Test too many paths that never used in functional mode?

Delay Testing Recommendation

- Must do **transition** delay fault ATPG
- Then select some critical paths for **PDF** ATPG
 - ♦ Timing analyzer (STA) selected critical paths
 - * e.g. >95% clock period
 - ♦ Designers' choice
 - * Remove special paths: e.g. multi-cycle paths
- If there are too many paths
 - ♦ Choose random samples that are not overlapped
- If they are not testable
 - ♦ Increase sample size or reduce threshold

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